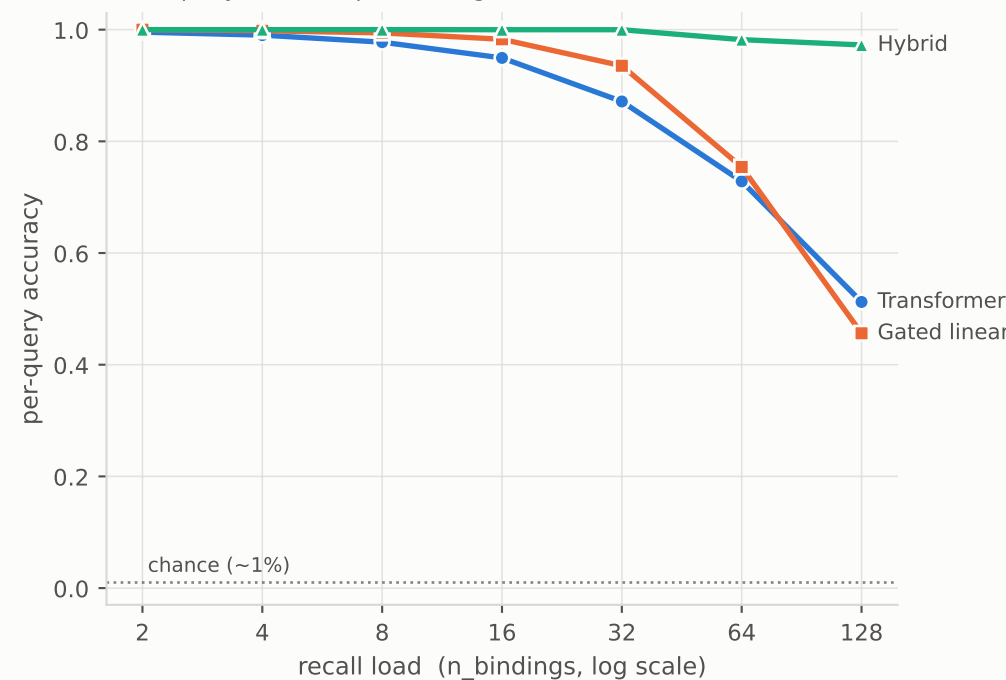


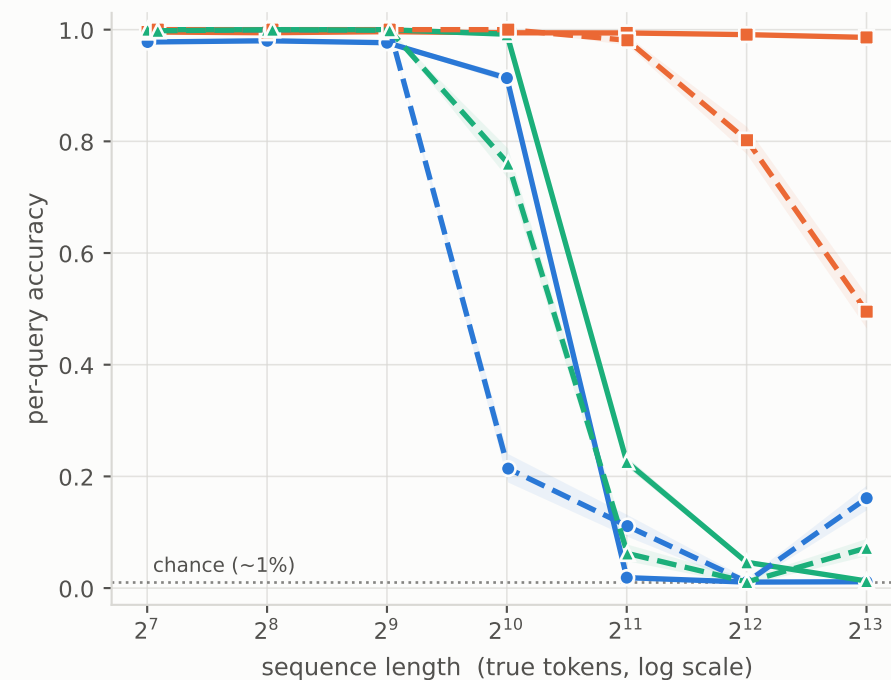
A • Capacity

multi-query recall, sequence length held in-distribution



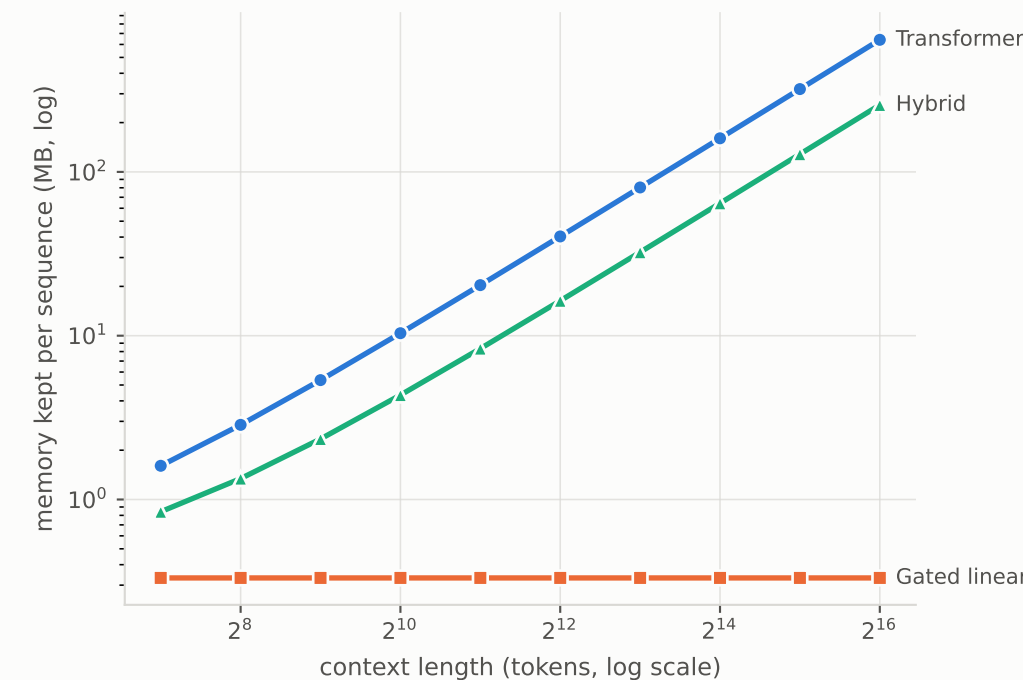
B • Length generalisation

solid: associative recall · dashed: state-track control



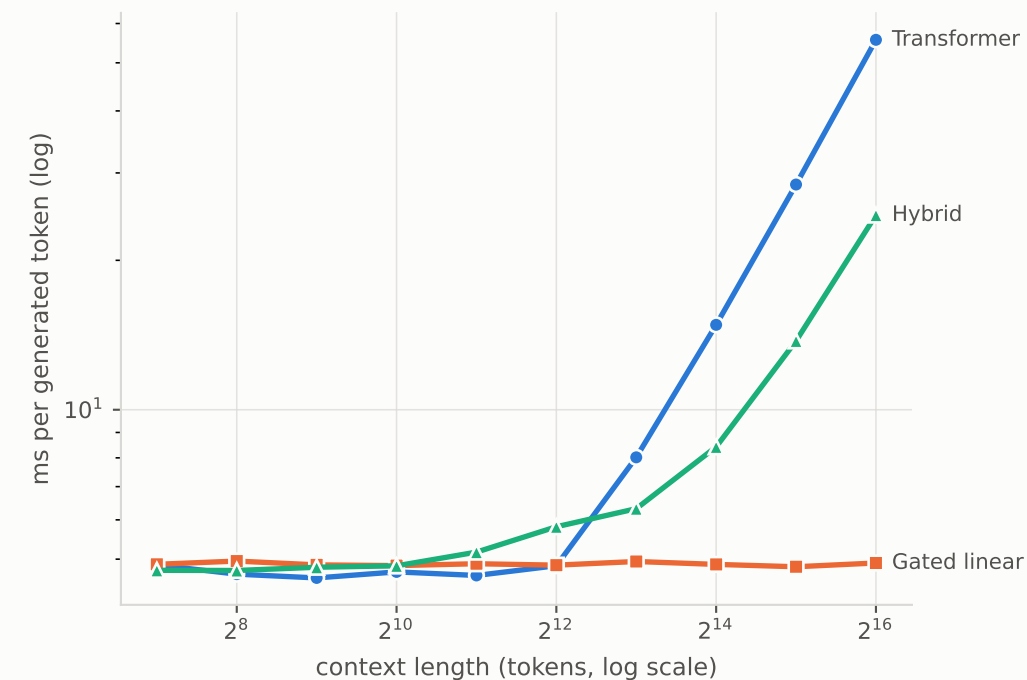
C • Decode memory

KV cache vs fixed state, batch 1



D • Decode latency

one token at a time, batch 1



Transformer (dense) Gated linear (fixed state) Hybrid